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PRELIMINARY PALEOMAGNETIC RESULTS FROM THE TRIASSIC PRIDA FORMATION (LOWER STAR PEAK GROUP), WEST-CENTRAL NEVADA, AND IMPLICATIONS FOR LATITUDINAL DISPLACEMENT

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Abstract. Paleomagnetic data were obtained from two oppositely dipping carbonate sections of the Triassic Prida Formation, lower Star Peak Group, in the Humboldt Range of west-central Nevada. Progressive demagnetization of the rocks indicated that the remanent magnetization consists of three or fewer vector components. A characteristic component, with the highest unblocking temperatures, was most effectively isolated by low temperature (200-300°C) thermal followed by progressive alternating field demagnetization. The sections yield in situ mean directions that differ significantly, but that have similar inclinations (Star Canyon: D=311, I=+34, α95=5, k=22, N= 38 samples; Fossil Hill: D=281, I=+33, α95=2°, k=2814, N= 3 section means, 154 samples). The directions diverge upon simple structural correction, from which we infer that at least one and probably both of the sections were remagnetized, probably in latest Middle to Late Jurassic time. These data cannot be used to indicate significant southward translation of Golconda allochthon rocks, as inferred from magnetizations in underlying volcanic rocks. The translation interpretation may require revision if magnetizations in the volcanic rocks are also secondary.

Introduction

In northwestern Nevada, late Paleozoic basinal rocks unconformably overlain by a lower Mesozoic volcanic, siliciclastic, and carbonate succession record two periods of tectonism associated with emplacement of the Permo-Triassic Golconda and late Mesozoic Fencemaker allochthons (Oldow, 1984; Speed et al., 1988). Basinal rocks of the Golconda allochthon were displaced eastward over the late Paleozoic to early Mesozoic continental margin during the Sonoma orogeny (Silberling, 1975; Speed et al., 1988) (Figure 1). Rocks as young as Guadelupian (Kazanian) are deformed within this allochthon and are overlain unconformably by a Triassic cover sequence composed of the volcanic Koipato Group and overlying carbonate and siliciclastic strata of the Star Peak Group (Silberling and Wallace, 1969; Nichols and Silberling, 1977). In late Middle Jurassic to mid-Cretaceous time, the Fencemaker thrust system carried lower Mesozoic basinal rocks southeasterly over the partially coeval Star Peak and Koipato Groups and the underlying Golconda allochthon (Oldow, 1984; Speed et al., 1988; Oldow et al., 1990).

The tectonic transport history of the Golconda allochthon and its cover is important in the pre-Tertiary development of the western Great Basin. On the basis of paleomagnetic data from 11 Koipato volcanic flows in the southern Tobin Range (Figure 1), Skalbeck et al. (1989) inferred that the Golconda allochthon was transported southward (about 1400 km) and rotated clockwise (about 25°), before Middle Tertiary time. They argued, principally on the basis of a positive fold test and dual polarities of magnetization, that the magnetization is primary and of latest Permian to Early Triassic age.

Secondary magnetizations, however, depending on the age and duration of magnetization acquisition, may yield positive fold tests; a positive fold test shows only that the magnetization antedates the folding. Remagnetization can also yield dual polarities (e.g., Van Alstine and Gillett, 1979; Gillett and Taylor, 1986). Similarity between the corrected Koipato magnetization (D= 350, I= +39) and expected mid-Jurassic directions, together with high (3-5+) conodont color alteration indices, corresponding to temperatures up to 300°C, in much of the area (Harris et al., 1980), suggest that Golconda allochthon and cover rocks may have been remagnetized. Paleomagnetic data obtained from Prida Formation carbonate strata, lower Star Peak

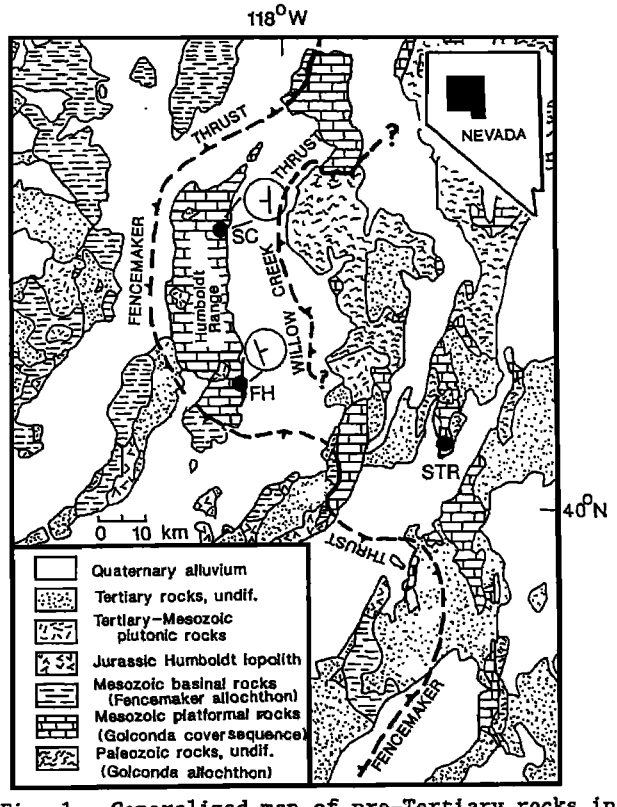


Fig. 1. Generalized map of pre-Tertiary rocks in western Nevada, modified from Oldow et al. (1990) and Speed et al. (1988). SC=Star Canyon; FH=Fossil Hill; STR=southern Tobin Range.

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Group, from two localities in the Humboldt Range, approximately 50 km west of the Tobin Range (Figure 1) support this suspicion.

Geology, Sampling, and Methodology

The stratigraphy, depositional history, and paleontology of the Star Peak Group have been described in detail (Silberling and Wallace, 1969; Nichols and Silberling, 1977). In the Prida Formation, locally abundant fossils are of late Anisian (Middle Triassic) age.

Two sections were sampled in 1975, originally for magnetostratigraphy, by the second author and E.M. Shoemaker. At Star Canyon (Figure 1), only the lower part of the Prida Formation (mean bedding attitude: 176/28) was sampled (165 samples). At Fossil Hill (Figure 1), the entire Prida section, arbitrarily divided into three members (mean bedding attitude: 340/20) was sampled (523 samples). At both localities, Star Peak strata overlie Koipato volcanic rocks with angular unconformity. Also, both localities occur in a zone of relatively little penetrative deformation related to the Fencemaker Thrust (Speed et al., 1989). Over 90 percent of the collection remained unanalyzed until 1988.

All specimens were rinsed for approximately 30 seconds in dilute HCl before recent measurements. Measurements of natural remanent magnetization (NRM) were made in a magnetically shielded room using a 3-axis superconducting rock magnetometer (model 760R from 2G Enterprises). Demagnetization employed commercial instrumentation. Directions of the remanent magnetization components were determined using principal component analysis using at least three and usually 5-6 steps; the origin was rarely used as an additional point. Approximately 40 percent of the original collection has been demagnetized.

Paleomagnetic Data

NRM intensities are typically around 1 mA/m (Table 1). NRM directions range from northerly declination and moderate positive inclination to westward and shallow positive values (Figure 2). Where the northward, positive NRM direction predominated, AF demagnetization rarely removed this magnetization, whereas thermal treatment usually isolated a second component of magnetization with a west to northwest declination and shallow positive inclination (Figure 2b). A component with east to southeast declination and shallow negative inclination, apparently reflecting reversed polarity, was also present in some samples (Figure 2d). The unblocking temperature for this component is between approximately 275 and 400°C. A useful demagnetization procedure proved to be heating to 200°C to 300°C to remove a high-coercivity component, probably residing in goethite or very fine-grained hematite, followed by AF

TABLE 1. Paleomagnetic Data, Prida Formation

Locality	In Situ N/No	(Nor.)	Dec.	Inc.	α95	k	Corr.1 Dec.	Corr.1 Inc.	Corr.2 Dec.	Corr.2 Inc.
Fossil Hill							<342,21>		<30,10>	
Upper	51/56	(41)	281	33	3	36	292	50	284	43
normal:	41		281	35	4	44				
reversed:	10		101	-28	10	23				
(mean NRM int. = 1.09mA/m, SD+ = 3.26, SD- = 0.36, n = 89)										
Middle	50/76	(49)	281	35	3	51	292	52	284	44
(mean NRM int. = 3.04mA/m, SD+ = 7.01, SD- = 1.32, n = 107)										
Lower	53/73	(50)	281	32	5	18	292	49	284	42
normal:	50		280	33	5	18				
reversed:	3		112	-17	-25	24				
(mean NRM int. = 1.28mA/m, SD+ = 2.86, SD- = 0.58, n = 231)										
Star Canyon							<206,26>		<30,10>	
Lower	38/58	(29)	311	34	5	22	304	14	317	41
normal:	29		311	37	5	25				
reversed:	9		131	-27	13	16				
(mean NRM int. = 0.82mA/m, SD+ = 1.84, SD- = 0.37, n = 166)										
Koipato Volcanics	11*/?	(5)	349	9	8	33	350	39	-	-
Humboldt lopolith extrusive rocks	9*/9*	(8)	-	-	8	28	280	52	-	-

Key N:number of samples(one specimen/sample demagnetized) accepted for statistical purposes (The somewhat high rejection rate is due to the fact that specimens demagnetized prior to recent work were not first thermally demagnetized at low temperatures), * site means; No:number of samples measured; Nor.:number of normal polarity samples out of N; Dec.,Inc.: declination and inclination; α95:semi-axis of the cone of P=0.05 confidence about mean direction; k=precision parameter; Corr.1: corrected directions, for Prida results, first using attitude of Miocene basalts, then of carbonate strata; Corr.2: partially corrected, using only attitude of Miocene basalts; values of corrections given in <>. Statistics for corrected Koipato data of Skalbeck et al. (1988) and corrected Humboldt lopolith extrusive data of Hudson (1988). SD+/SD-: Standard deviations of a log normal distribution, n: number of individual specimens whose NRM was measured.

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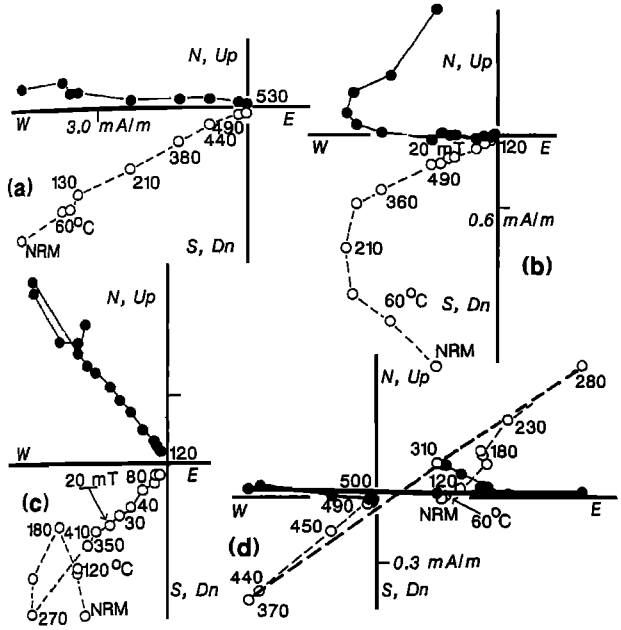


Fig. 2. Examples of response to progressive demagnetization by Prida Formation rocks. Orthogonal demagnetization diagrams, in situ coordinates. (a) Specimen 84.001, lower Prida, Fossil Hill; thermal demagnetization to 530°C; (b) Specimen 165.001, lower Prida, Fossil Hill; thermal to 490°C, followed by AF from 20 to 120 mT; (c) Specimen 317.001, lower Prida, Star Canyon; thermal to 410°C, followed by AF from 20 to 80 mT; (d) Specimen 60.001, upper Prida, Fossil Hill; thermal to 500°C; below 280°C, a normal polarity component is isolated, from 280 to 370°C, a reverse polarity, and above 370°C, a normal polarity component.

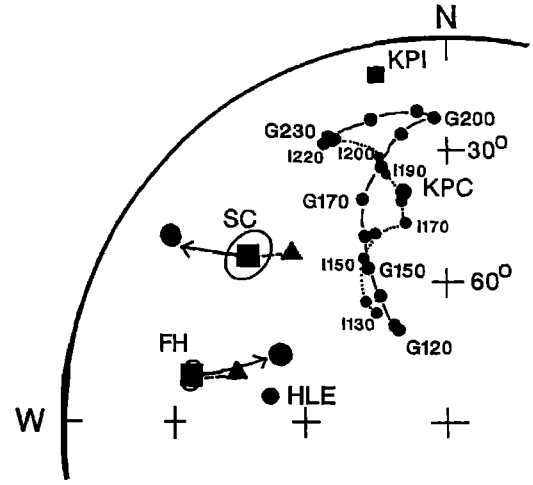


Fig. 3. Equal area projection of locality mean paleomagnetic data, Prida Formation. Squares refer to in situ means, circles to corrected data (Corr.1, Table 1), triangles to corrected data (Corr.2, Table 1). FH=Fossil Hill, SC=Star Canyon, KPI=Koipato (in situ), KPC=Koipato(corr), HLE=Humboldt Lopolith extrusives (corr). Koipato data from Skalbeck et al. (1989); Humboldt lopolith data from Hudson (1988). Expected directions for Mesozoic time periods based on paleomagnetic poles of Irving and Irving (1982) (I) and Gordon et al. (1984)(G), respectively.

demagnetization to levels greater than 100 mT (Figure 2b,c). Even for samples with the high-coercivity component, IRM acquisition curves documented the abundance of a magnetic phase whose magnetization saturated below about 200 mT, probably magnetite.

Both sections yielded well-grouped directions containing both polarities, which are antiparallel at the 95 percent confidence level (Table 1). The reversed polarity direction is poorly constrained, however, owing to the small number of samples. Section mean directions are compared with Mesozoic expected directions in Figure 3. The in situ Star Canyon and Fossil Hill means, although statistically indistinguishable in inclination, differ significantly in declination (~30°). As the sections dip in opposite directions, correcting for bedding tilt causes the inclinations to diverge (Table 1, Figure 3), providing evidence for a negative but statistically undefined (N=2) fold test.

Discussion and Conclusions

One and probably both of the Prida Formation sections were remagnetized. We argue that the remagnetization reflects tectonothermal events in latest Middle to early Late Jurassic (ca. 170–150 Ma) time, possibly during subsidence of shelfal units accompanying initial stages of thrust loading in southeastward transport along the Fencemaker thrust. In addition, at approximately 160 Ma (Hudson, 1988), the area was intruded by a large mafic complex, the "Humboldt lopolith", as well as by granitic plutons spanning the 170 to 150 Ma period. Although we prefer not to make a direct association between lopolith emplacement and remagnetization, we note that the in situ, as well as partially corrected (see below), Prida results are similar to structurally corrected data from extrusive rocks of the lopolith (Table 1, Figure 3) which are predominantly of normal polarity and appear to have been magnetized penecontemporaneously (Hudson, 1988). The counterclockwise discordance between in situ and expected declinations may be a product of both Mesozoic and Cenozoic phases of deformation. Part of the discordance may have resulted from southeastward contraction in post mid-Jurassic and pre-Late Cretaceous time. This phase of deformation possibly contributed to the arcuate, convex westward shape of the Humboldt Range. However, effects of Cenozoic extension in this vicinity must also be considered, but are difficult to quantify because of the lack of Cenozoic cover for paleomagnetic study. The discrepancy in declination between the in situ Fossil Hill and Star Canyon directions (Figure 3) is consistent with Cenozoic counterclockwise rotation of parts of the southern Humboldt Range, similar to that which affected the northern Dixie Valley region in middle Tertiary time (Hudson and Geissman, 1987).

Along the southeastern margin of the Humboldt Range, Miocene basalt flows dip ~10° to the east-northeast. If this tilt alone is removed from the in situ means at both localities, the inclinations steepen and agree even better with expected Late Jurassic values (Figure 3). Although mid-Tertiary tilting may exist, it is difficult to evaluate from available exposures. If mid-Tertiary deformation involved an even greater amount of eastward tilt, the corrected Prida data would have even steeper inclinations, implying a possible association between remagnetization and mid-Cretaceous plutonism and mineralization in parts of the Humboldt Range. Paleomagnetic data from the Prida Formation indicate that remagnetization occurred after some degree of deformation, arguably during early stages (first phase, Oldow et al., 1990) of contraction along the Fencemaker thrust system in Late Jurassic time. Remagnetization may have resulted from thermal resetting, a possibility implied by the high color alteration indices in the region. Alternatively, remagnetization may have resulted from magnetite authigenesis by regional, pervasive hydrothermal activity associated with the Humboldt lopolith. Battles and Barton (1989) have

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shown that Na-Ca metasomatism associated with Fe mineralization is pervasive in the lopolith itself. Skalbeck et al. (1989) noted that their results could be explained by remagnetization during the Late Triassic to Late Cretaceous, but they argued for a southward translation explanation. To explain our data as well as the Koipato results, we suggest that the Prida Formation and underlying volcanic rocks were remagnetized in Late Jurassic time. Southeastward progression of the Fencemaker thrust and upper plate rocks could have subsequently folded the Koipato volcanic rocks.

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